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SCHOLARLY REVIEW

Inside the Belly of a Beast: Individualizing Nutrition for Young, Professional Male Rugby League Players: A Review

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Professional rugby league (RL) football is a contact sport involving repeated collisions and high-intensity efforts; both training and competition involve high energy expenditure. The present review summarizes and critiques the available literature relating the physiological demands of RL to nutritional requirements and considers potential ergogenic supplements that could improve players' physical capacity, health, and recovery during the preparatory and competition phases of a season. Although there may not be enough data to provide RL-specific recommendations, the available data suggest that players may require approximately 6–8 $g \cdot kg^{-1} \cdot day^{-1}$ carbohydrate, 1.6–2.6 $g \cdot kg^{-1} \cdot day^{-1}$ protein, and 0.7–2.2 $g \cdot kg^{-1} \cdot day^{-1}$ fat, provided that the latter also falls within 20–35% of total energy intake. Competition nutrition should maximize glycogen availability by consuming 1–4 g/kg carbohydrate (~80–320 g) plus 0.25 g/kg (~20–30 g) protein, 1–4 hr preexercise for 80–120 kg players. Carbohydrate intakes of approximately 80–180 g (1.0–1.5 g/kg) plus 20–67 g protein (0.25–0.55 g/kg) 0–2 hr postexercise will optimize glycogen resynthesis and muscle protein synthesis. Supplements that potentially improve performance, recovery, and adaptation include low to moderate dosages of caffeine (3–6 mg/kg) and ~300 mg polyphenols consumed ~1 hr preexercise, creatine monohydrate "loading" (0.3 $g \cdot kg^{-1} \cdot day^{-1}$) and/or maintenance (3–5 g/day), and beta-alanine (65–80 $mg \cdot kg^{-1} \cdot day^{-1}$). Future research should quantify energy expenditures in young, professional male RL players before constructing recommendations.

***Keywords**:* ergogenic aids, football, nutrient periodization, nutritional strategies, supplementation

Professional rugby league (RL) football is a high-intensity collision sport increasingly played by both males and females in Australia, New Zealand, the United Kingdom, France, and, recently, Canada. In Australia, a typical National Rugby League (NRL) season will cover approximately 10–11 months, including a general preparatory phase (e.g., November–February) and competitive phase (March–October) (Gabbett & Jenkins, 2011). In training and competition, players demonstrate high energy expenditures (Costello et al., 2018a), incur collision-induced muscle damage, and experience reduced markers of well-being (Bradley et al., 2016; Costello et al., 2018b). Recovery between training sessions optimizes health (complete physical, mental, and social well-being) (World Health Organization, 1995) and could enhance adaptations during both preseason, when training volumes are the greatest, and in-season, when intense training and competitive rounds coexist (Weaving et al., 2019). Recently, recommendations for the periodization of nutrition practices for soccer over the pre- and in-season competition phases have been proposed (Ranchordas et al., 2017). While limited literature has explored preseason (Costello et al., 2018b) and prematch (Bradley et al., 2016) nutrition in RL, there are no nutritional recommendations across a complete season in RL. Although beyond the scope of this review, future research should also look to apply similar female-specific RL recommendations, and readers are directed to the review of Hausswirth and Le Meur (2011) for further understanding. Young, professional males often play RL at the under-16 (U16), under-20 (U20), and under-24 (U24) levels. This review therefore provides insights into the application of periodized nutrition for professional male RL players, aged approximately 16–24 years, based on the available research, to maximize recovery, adaptation, and performance.

Physiological Demands of RL

Professional RL involves two teams consisting of 13 players (six forwards and seven backs), contesting two 40-min halves interspersed by a 10-min half-time interval (Gabbett et al., 2008). Players carry approximately 80–120 kg of body mass (BM) across 4,000–7,500 m for backs and 3,500–6,000 m for forwards (Gabbett et al., 2012). Relative distance (total distance traveled per minute of match play; in meters per minute) quantifies match intensity as it considers on-field match time (Hausler et al., 2016). A systematic review and meta-analysis of studies including players aged

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approximately 15–24 years highlighted the large variability in match and training profiles of RL players; however, mean relative distances are highest in the elite NRL and European Super League (United Kingdom and France; 85–104 m/min), with lower intensities at the subelite (74–97 m/min) and junior (73–91 m/min) levels (Hausler et al., 2016). Players face frequent bouts of high-intensity activity (e.g., running and passing, sprinting, tackling) during game-based training (one bout every 4.5 min) and NRL competition (one bout every 6.9 min) (Gabbett et al., 2008, 2012). During an NRL match, players with a mean age of 23.6 years could compete in 23–52 collisions, where 9–18 are heavy clashes (contact made, momentum stopped, and player forced backwards; Gabbett et al., 2012). These data show that RL is a game that requires sustained high-intensity intermittent exercise.

Data from 16 academy RL players aged 17.2 ± 0.7 years suggested that players trained 88 times during a 14-week preseason, comprising 35 resistance training (40%), 30 rugby (34%), and 23 conditioning (26%) sessions (Dobbin et al., 2018). For elite players, training can involve an average of 77 collisions per player, per session across a season (Gabbett et al., 2010), and they may train twice daily on consecutive days (Tavares et al., 2017). Represented as the number of training sessions at different intensities (rest/light/heavy), U16 (8/3/4), U20 (5/2/7), and U24 (4/8-9/1) players have been shown to have intense training demands during a 14-day in-season period, in addition to 0–2 matches (Smith et al., 2018). The higher frequency of heavy sessions in U20s, which consisted of both resistance and rugby training sessions, may be explained by the ability to better handle concurrent training loads, as shown in college-aged RL players (18.2 ± 1.0 years old; Baker, 2001). In addition, given that the U24s had 0–2 matches, compared with one for the U20s, coaches may have prescribed fewer heavy sessions for the U24s to avoid the increased training injury rates associated with higher training loads (Gabbett, 2004). Nutrition recommendations should adapt to different frequencies and demands of training and match play to optimize players' performance.

Nutrition Periodization in Sport

Recently, sports nutrition research has advanced the idea of individualizing nutrition through the concept of "fuel for the work required" (Impey et al., 2018). This approach describes the notion of a glycogen threshold and periodization, adjusting carbohydrate (CHO) availability to optimize endurance adaptations. A recent publication addressed the implementation of periodized nutrition in athletics (Stellingwerff et al., 2019), and an improved understanding of periodized nutrition in elite athletes over prolonged training and competition phases is needed (Stellingwerff et al., 2019). Though practical recommendations for team sports have been offered (Heaton et al., 2017), nutritional advice, specifically for RL players and that which addresses the unique challenges of the game, is lacking.

Energy Requirements and Expenditures

Energy availability is the amount of dietary energy remaining for all other physiological systems after a given challenge (e.g., exercise) has been overcome and is normalized to fat-free mass (FFM), whereby >45 kcal·kg $FFM^{-1}\cdot day^{-1}$ may be optimal (Loucks, 2013). Importantly, maximizing muscle glycogen availability (Bradley et al., 2016) and postexercise muscle protein synthesis (MPS; Morton et al., 2015) likely optimizes recovery and performance capabilities. This should be approached in lieu of the immunosuppressive nature of RL as a consequence of intense training and competition (Cunniffe et al., 2011). Although FFM data was lacking, in-season estimates of resting metabolic rates across 1 week in elite rugby union (RU) players (age; 25.7 ± 4.1 years; mass: 104 ± 12.6 kg) using the doubly labeled water method have been reported as 2,313 ± 283 to 2,544 ± 397 kcal/day (approximately 22.2–24.5 $kcal\cdot kg^{-1}\cdot BM^{-1}$) (Hudson et al., 2020). In professional RL players weighing 85.0–103.9 kg (69–82 kg lean mass), estimated resting metabolic rate measures taken across two consecutive weeks in-season were 1,713–1,975 kcal/day (approximately 17.6–23.2 $kcal\cdot kg^{-1}\cdot BM^{-1}$) (Morehen et al., 2016). In six professional young (age; 17.2 ± 0.7 years) male RL players, total energy expenditures estimated by doubly labeled water were significantly higher (~239 kcal/day) during a 5-day, preseason collision-based training microcycle (~4541 ± 796 kcal/day; ~52 $kcal\cdot kg^{-1}\cdot BM^{-1}$) than during an identical noncollison microcycle (~4302 ± 811 kcal/day; 49 $kcal\cdot kg^{-1}\cdot BM^{-1}$) (Costello et al., 2018b). With FFM data available, average doubly labeled water–estimated total energy expenditures were 4,010 ± 744 (62 ± 8 kcal·kg $FFM^{-1}\cdot day^{-1}$), 4,414 ± 618 (66 ± 10 kcal·kg $FFM^{-1}\cdot day^{-1}$), and 4,761 ± 1,523 kcal/day (60 ± 18 kcal·kg $FFM^{-1}\cdot day^{-1}$) in-season for professional U16, U20, and U24 players, respectively (Smith et al., 2018). Effect sizes (ESs) showed *unclear* differences (confidence interval: 90%) between U16 and U20s (ES = −0.55; [−1.31, 0.22]) and U20 and U24s (ES = −0.19; [−1.03, 0.64]). Although unclear, these trends suggest that it is possible that players' energy requirements increase from the academy to senior level, exacerbating the already increased demands for growth and maturation. Indeed, the average resting metabolic rate in 18 ± 1-year-old male RL players was ~2,346 kcal/day the day before a match and increased significantly both 24 hr (~3,007 kcal/day; ~568 kcal higher) and 48 hr after the match (~2,782 kcal/day; ~436 kcal higher) (Costello et al., 2019). The emerging data on variable energy requirements allow practical applications to be proposed (Costello et al., 2019). There was no clear effect of increasing meal frequency from four to six meals per day on the lean mass of developing elite RU and RL players (age = 20.1 ± 1.4 years) who reportedly consumed 1.3 ± 0.4 $g\cdot kg^{-1}\cdot day^{-1}$ fat, 2.7 ± 0.6 $g\cdot kg^{-1}\cdot day^{-1}$ protein, and 3.6 ± 1.3 $g\cdot kg^{-1}\cdot day^{-1}$ CHO (3,771 ± 879 to 3,773 ± 785 kcal/day) (MacKenzie-Shalders et al., 2016). Akin to the athletes in this study, young male RL players should focus on evenly distributing protein intakes of 2.0–2.6 $g\cdot kg^{-1}\cdot day^{-1}$, possibly including supplements after training. Therefore, suggestions of how general athlete guidelines (Burke et al., 2011; Thomas et al., 2016) may translate to RL (Costello et al., 2019) and RU players' requirements (Hudson et al., 2020) are shown in Tables 1–6.

Carbohydrate Periodization

Preparatory and Competition Phase Requirements

The CHO-dependent energy systems (aerobic and anaerobic) are used in RL (Gabbett, 2004), and muscle glycogen has been shown to decrease by ~40% during academy matches (Bradley et al., 2016). Maximizing muscle glycogen preexercise through CHO intakes of 6–10 $g\cdot kg^{-1}\cdot day^{-1}$ is therefore crucial (Burke et al., 2017), the periodization of which is specific to each sport's traditional practices (Impey et al., 2018). However, this raises questions as to whether 10 $g\cdot kg^{-1}\cdot day^{-1}$ (>1,000 g CHO for >100 kg players) is realistically achievable. Young players have

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Table 1 Suggested Nutritional Strategies to Optimize Performance and Recovery in Rugby League Players

Nutrition intervention	Prematch recommendation	Timing	Notes	Reference	Postmatch recommendation	Timing	Notes	Reference
Carbohydrate	6–8 g/kg	36 hr prematch	1.5–2.0 $g \cdot kg^{-1} \cdot BM^{-1}$ per meal in 3–6 meals	Bradley et al. (2016)	80–180 g	0–1 hr postmatch Resume normal eating habits thereafter	Palatable food and drink provided in dressing rooms or take-home packs	Burke et al. (2016)
	1–4 g/kg	1–4 hr prematch	The closer to kick-off, the lower in fiber (i.e., high GI)	Thomas et al. (2016)				
	20–30 g	30–60 min prematch	Consume light snack/drink after warm-up, if hungry	Thomas et al. (2016)				
	30–60 g	During match	If approximately 60 min played	Burke et al. (2011)				
Protein	1.6–2.6 $g \cdot kg^{-1} \cdot day^{-1}$		0.25–0.4 g/kg high-quality sources distributed evenly across 3–6 meals	Schoenfeld and Aragon (2018); Thomas et al. (2016)	20–67 g	0–1 hr postmatch	Palatable, high-quality sources provided in dressing rooms or take-home packs	Kerksick et al. (2017)
					30–40 g casein	30–60 min before bed	May improve sleep quality and/or quantity	Jäger et al. (2017)
Hydration	5–7 ml/kg	2–4 hr prematch	Consume approximately 400–700 ml, achieving pale yellow urine before kick-off	Sawka et al. (2007)	1.25–1.5 L/kg of BM lost	As soon as practically possible	Combine fluid and food intake to restore fluid balance, for e.g., in dressing rooms and take-home packs	Ranchordas et al. (2017)
Whey protein					20–40 g powder	0–1 hr postmatch	Muscle most sensitive to protein ingestion immediately postexercise	Morton et al. (2015)
Casein					30–40 g powder	30–60 min before bed	May improve sleep quality and/or quantity	Jäger et al. (2017)
Creatine monohydrate	3–5 g/day or 0.3 $g \cdot kg^{-1} \cdot day^{-1}$ for 5 days plus 3–5 g/day thereafter	1 × 3–5 g dosage 4 × 5–6 g dosage	Weight gain possible due to water retention Consume in equal amounts throughout the day	Kreider et al. (2017)				

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Table 1 *(continued)*

Nutrition intervention	Prematch recommendation	Timing	Notes	Reference	Postmatch recommendation	Timing	Notes	Reference
Caffeine	3–6 mg/kg	1 hr prematch	Approximately 240–600 mg (e.g., 3 × 100 mg or 2 × 200 mg tablets/capsules)	Goldstein et al. (2010)				
		Consider individual response (e.g., if hyper/unresponsive)	Possible side effects include disrupted sleep and jitters	Burke (2008); Halson (2014)				
β-Alanine	4–6 g/day for 4 weeks (approximately 65–80 $mg \cdot kg^{-1} \cdot day^{-1}$)	Consume lower doses throughout the day	Reduces likelihood of paraesthesia (i.e., tingling)	Kelly et al. (2013); Trexler et al. (2015)				
Polyphenols	300 mg from fruit	1 hr prematch (e.g., 300 g montmorency cherries, 450 g blueberries, or 120 g blackcurrants)	May improve repeated sprint performance	Bowtell and Kelly (2019)	30 ml Montmorency cherry concentrate	One of dosages 30–60 min before bed	May improve sleep quality and/or quantity	Halson (2014); Howatson et al. (2012)
	>1,000 mg/day supplementation (e.g., pomegranate, blackcurrant, blueberry extracts or 2 × 30 ml/day Montmorency cherry concentrate)	Consistently for 3 days prior	May improve muscle damage recovery	Bowtell and Kelly (2019)				

Note. BM = body mass; GI = glycemic index.

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Table 2 Daily Carbohydrate Periodization for a 104-kg Player (Mean Age; 25.7 ± 4.1 Years) Aiming to *Lose* 0.5-kg Body Mass Per Week In-Season (500-kcal Deficit/Day) Based on the RMR Reported by Hudson et al. (2020) With Physical Activity Levels of 1.4 as "Light" (e.g., Sedentary/Seated Work; Black, 2001)

Diet and activity	MD −1	MD +1	MD +2
Meal 1 (07:30 a.m.)	Large bowl porridge (80 g dry oats) made with 250 ml semiskimmed milk, 2 tbsp sunflower and pumpkin seed mix, one small banana, and 150 g 0% fat plain yogurt. 0.5 L water. Extra 0.5 L water throughout morning.	Large bowl porridge (80 g dry oats) made with 375 ml semiskimmed milk, one medium banana, 125 g 0% fat plain yogurt, 3 tbsp honey, and 1 tbsp peanut butter. 0.5 L water. Extra 0.5 L water throughout morning.	Large bowl muesli (149 g) made with 250 ml semiskimmed milk and 2 tbsp sunflower and pumpkin seed mix with a protein shake (1 pint semiskimmed milk and one scoop flavored whey protein) with one large orange. 0.75 L water.
Meal 1 code	LOWER	HIGHER	MOD
Meal 2 (12:30 p.m.)	**Smoothie—1 pint semiskimmed milk, one scoop whey protein, 1 tbsp peanut butter, and 160 g tropical smoothie frozen fruit mix.** **kcal: 577** **46 g PRO (0.4 $g^{-1}\cdot kg^{-1}$)** **49 g CHO (0.5 $g^{-1}\cdot kg^{-1}$)** **22 g FAT (0.2 $g^{-1}\cdot kg^{-1}$)**	Chicken, bacon, and mushrooms cooked in a sauce with 250 g Mediterranean vegetables (½ 500 g bag) with a mini bruschetta and red pepper topping. 0.5 L water.	125 g (small can) mackerel in tomato sauce, 84 g asparagus, 1 cup cherry tomatoes, and two medium slices brown toast. 0.75 L water.
Meal 2 code	LOWER	LOWER	MOD
PM training (13:30 pm)			*Resistance training and light mobility (1 hr)* *0.5 L water throughout training.*
Meal 3 (15:30 p.m.)	Two tuna wraps, each made up of wheat flour tortilla, 80 g mixed salad, 1 tbsp low-fat mayonnaise, 40 g sweet corn, 40 g mixed salad, and ½ can tuna. 0.5 L water.	Smoothie—1 pint semiskimmed milk, one scoop whey protein, ½ cup chopped dates, 80 g blueberries, 3 tbsp honey, 1 cup strawberries halved, and Nakd bar. 0.5 L water before next meal.	**Roast dinner: one chicken leg with skin, 150 g gravy, one Yorkshire pudding, 90 g carrots, 3 tbsp peas, small portion roast potatoes, medium portion broccoli, cabbage, 4 tbsp stuffing, and 2 tbsp Bramley apple sauce. 0.5 L water.** **kcal: 980** **80 g PRO (0.8 $g^{-1}\cdot kg^{-1}$)** **81 g CHO (0.8 $g^{-1}\cdot kg^{-1}$)** **37 g FAT (0.4 $g^{-1}\cdot kg^{-1}$)**
Meal 3 code	LOWER	HIGHER	MOD
Meal 4 (19:00–20:00 p.m.)	Medium portion (160 g) white rice, 125 g bean sprouts, 80 g stir-fry vegetables pan-fried in olive oil, medium portion (140 g) extra lean beef mince with 200 g Chinese stir-fry sauce, and large portion broccoli (110 g). 0.75 L water.	Flavored protein shake (one scoop whey protein) with water, 1 cup of mixed berries, and 2 × soya yogurt pots with a cup of hot chocolate.	One hot cross bun with 1.5 tbsp butter. 0.5 L water before bed.
Meal 4 code	MOD	LOWER	LOWER
Daily nutrition profile	kcal: 2,728 (RMR 2,313 × PAL 1.4 = 3,238 − 500 = 2,728) CHO 3.1 $g^{-1}\cdot kg^{-1}$, PRO 1.7 $g^{-1}\cdot kg^{-1}$, FAT 1.3 $g^{-1}\cdot kg^{-1}$ (322 g) (178 g) (81 g)	kcal: 3,077 (RMR 2,544 × PAL 1.4 = 3,562 − 500 = 3,062) CHO 3.6 $g^{-1}\cdot kg^{-1}$, PRO 1.9 $g^{-1}\cdot kg^{-1}$, FAT 0.8 $g^{-1}\cdot kg^{-1}$ (375 g) (198 g) (87 g)	kcal: 2,853 (RMR 2,391 × PAL 1.4 = 3,347 − 500 = 2,847) CHO 2.6 $g^{-1}\cdot kg^{-1}$, PRO 1.8 $g^{-1}\cdot kg^{-1}$, FAT 1.1 $g^{-1}\cdot kg^{-1}$ (272 g) (184 g) (114 g)

Key		Meal code	Carbohydrate content
MD −1	The day before match day	HIGHER	Higher: 1.2–1.5+ g·kg
MD +1	The day after match day	MOD	Moderate: 1.0–1.2 g·kg
MD 2	Two days after match day	LOWER	Lower: 0.5–1.0 g·kg

Note. Fluid intake meets ~1 ml/kcal energy expenditure. These values are seen as bare minimum to encourage adequate energy intake and are not recommendations, as, for example, 3.1 g/kg CHO on MD −1 is unlikely to maximize glycogen stores. For an example of what players may be required to eat around match day (6–8 g/kg), see Table 6. Meals highlighted in **bold** show examples of the nutritional contents of meals. Fluid requirements may fluctuate depending on training responses, for example, sweat rate. Differences in meals and codes are simply to account for appetite variation. Estimated using Nutritics (2019). Italics indicate training activities and dietary intakes. CHO = carbohydrate; FAT = dietary fat; MD = match day; PAL = physical activity level; PM = afternoon; PRO = protein; RMR = resting metabolic rate; tbsp = tablespoon.

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Table 3 A Similar Diet for the Same 104-kg Player (Mean Age; 25.7 ± 4.1 Years) Aiming to *Gain* 0.5-kg Body Mass Per Week In-Season (500 kcal Surplus/Day) Based on the RMR Reported by Hudson et al. (2020) With Physical Activity Levels of 1.4 as "Light" (e.g., Sedentary/Seated Work; Black, 2001)

Diet and activity	MD -1	MD +1	MD +2
Meal 1 (07:30 a.m.)	Omelette (four eggs), 85 g spinach, and one medium tomato. Two crumpets (1 tsp butter each) and 140 g fruit salad. 0.5 L water. 0.5 L extra water throughout morning.	Omelette (two eggs), 85 g spinach, and one medium tomato. Two crumpets (1 tsp butter each) and 140 g fruit salad. 0.5 L water. 0.5 L extra water throughout morning.	Very large bowl muesli (149 g) made with 250 ml semiskimmed milk and 10 g sunflower and pumpkin seed mix with a protein shake (1 pint semiskimmed milk and one scoop whey protein). One slice fruited malt bread with one large orange. 0.5 L water.
Meal 1 code	LOWER	LOWER	HIGHER
Meal 2 (12:30 p.m.)	Two sandwiches, each with: two slices deli turkey, ½ salad tomato, small handful lettuce, and 1 tsp unsalted butter. 0.5 L water.	Two sandwiches, each with: two slices deli turkey, ½ salad tomato, small handful lettuce, and 1 tsp unsalted butter. 0.5 L water.	125 g (small can) mackerel in tomato sauce; 84 g asparagus; 2 cups cherry tomatoes; two medium slices brown toast; and two hot cross buns, each with 1 tsp butter. 0.75 L water.
Meal 2 code	LOWER	MOD	HIGHER
PM training (14:30 p.m.)			*Resistance training and light mobility (1 hr)* *0.75 L water throughout training.*
Meal 3 (16:00 p.m.)	Large (426 g) Tuscan pasta bake and 80 g courgette with 250 ml fruit juice. 0.75 L water.	Large (426 g) Tuscan pasta bake and 80 g courgette with 250 ml fruit juice. 0.75 L water.	Two slices wholemeal toast with 300 g cottage cheese. One bag Propercorn sweet and salty popcorn. One small-medium orange. 0.75 L water.
Meal 3 code	MOD	MOD	LOWER
Meal 4 (19:00 p.m.)	**160 g easy-cook brown rice (cooked), 160 g chicken curry with vegetables and curry sauce, one medium bell pepper, 80 g spring onions, and ½ plain paratha flatbread. 0.75 L water.** **kcal: 692** **35 g PRO (0.3 $g^{-1}\cdot kg^{-1}$)** **95 g CHO (0.9 $g^{-1}\cdot kg^{-1}$)—close to "amber"** **18 g FAT (0.2 $g^{-1}\cdot kg^{-1}$)**	160 g easy-cook brown rice (cooked), 160 g chicken curry with vegetables and curry sauce, ½ medium bell pepper, 80 g spring onions, and ½ plain paratha flatbread. 0.75 L water.	**One medium (grilled) skinless chicken breast, one large baked sweet potato, 1 cup vegetable stir-fry (fried in olive oil), and 1 × 75 g focaccia herb/garlic bread. 0.5 L water.** **0.75 L extra water before bed.** **kcal: 967** **65 g PRO (0.6 $g^{-1}\cdot kg^{-1}$)** **136 g CHO (1.3 $g^{-1}\cdot kg^{-1}$)** **18 g FAT (0.2 $g^{-1}\cdot kg^{-1}$)**
Meal 4 code	LOWER	MOD	HIGHER
Meal 5 (20:30 p.m.)	Two white bagels, each with: 112 g smoked salmon, 1 tbsp extra light cream cheese, and ¾ cup cucumber slices; and Nakd bar. 0.75 L water.	Two white bagels, each with: 112 g smoked salmon, 1 tbsp extra light cream cheese, and ¾ cup cucumber slices; and Nakd bar. 0.75 L water.	
Meal 5 code	HIGHER	HIGHER	
Meal 6 (22:00 p.m.)		250 g quark, 1 cup strawberry halves, 3 tbsp honey, and 25 g dark chocolate. 0.75 L water.	
Meal 6 code		LOWER	
Daily nutrition profile	kcal: 3,746 (RMR 2,313 × PAL 1.4 = 3,238 + 500 = 3,738) CHO 4.1 $g^{-1}\cdot kg^{-1}$, PRO 1.9 $g^{-1}\cdot kg^{-1}$, FAT 1.2 $g^{-1}\cdot kg^{-1}$ (429 g) (201 g) (132 g)	kcal: 4,067 (RMR 2,544 × PAL 1.4 = 3,562 + 500 = 4,062) CHO 4.8 $g^{-1}\cdot kg^{-1}$, PRO 2.1 $g^{-1}\cdot kg^{-1}$, FAT 1.2 $g^{-1}\cdot kg^{-1}$ (501 g) (223 g) (126 g)	kcal: 3,859 (RMR 2,391 × PAL 1.4 = 3,347 + 500 = 3,847) CHO 4.8 $g^{-1}\cdot kg^{-1}$, PRO 2.2 $g^{-1}\cdot kg^{-1}$, FAT 1.0 $g^{-1}\cdot kg^{-1}$ (500 g) (225 g) (108 g)

Key		Meal code	Carbohydrate content
MD −1	The day before match day	HIGHER	Higher: 1.2–1.5+ g·kg
MD +1	The day after match day	MOD	Moderate: 1.0–1.2 g·kg
MD 2	Two days after match day	LOWER	Lower: 0.5–1.0 g·kg

Note. Foods in **bold** show portion differences and additions. Meals highlighted in bold show examples of the nutritional contents of meals. Fluid requirements may fluctuate depending on training responses, for example, sweat rate. Differences in meals and codes account for appetite variation. Estimated using Nutritics (2019). CHO = carbohydrate; FAT = dietary fat; PAL = physical activity level; PM = afternoon; PRO = protein; RMR = resting metabolic rate; tbsp = tablespoon; tsp = teaspoon.

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Table 4 Daily Carbohydrate Periodization for a 16- to 18-Year-Old, 92.4-kg Male RL Player Aiming to *Maintain* Body Mass In-Season Based on the Estimated RMR of Costello et al. (2019) With Physical Activity Levels of 1.4 as "Light" (e.g., Sedentary/Seated Work; Black, 2001)

Diet and activity	MD 0	MD +1	MD +2
Meal 1 (09:00 a.m.)	Very large bowl porridge (1.5 cups dry oats) made with ¾ pint semiskimmed milk, one medium banana, 1 tbsp peanut butter, 2 tbsp honey, and 150 g 0% fat plain yogurt. 0.5 L water.	Very large bowl porridge (1.5 cups dry oats) made with 375 ml semiskimmed milk, one medium banana, 2 tbsp peanut butter, and 125 g 0% fat plain yogurt. 0.5 L water. Extra 0.5 L throughout morning.	Medium bowl muesli (149 g) made with 250 ml semiskimmed milk and 1.5 tbsp sunflower and pumpkin seed mix with one large orange. 0.5 L water.
Meal 1 code	HIGHER	HIGHER	LOWER
Meal 2 (13:00 p.m.)	Two tuna wraps, each with: wholemeal tortilla, 80 g mixed salad, 1 tbsp low-fat mayonnaise, 3 heaped tbsp sweet corn, and ½ can tuna; and Nakd bar. 0.5 L water.	**Four medium slices brown toast, one can baked beans, 10 g low-fat spread, three eggs, and Nakd bar. 0.75 L water. kcal: 1,079 65 g PRO (0.7 $g^{-1}\cdot kg^{-1}$) 129 g CHO (1.4 $g^{-1}\cdot kg^{-1}$) 34 g FAT (0.4 $g^{-1}\cdot kg^{-1}$)**	125 g (small can) mackerel in tomato sauce; 84 g asparagus; 2 cups cherry tomatoes; two medium slices brown toast; and two hot cross buns, each with 1 tsp butter. 0.75 L water.
Meal 2 code	HIGHER	HIGHER	HIGHER
PM match/ training (15:00 p.m.)	*Match (warm-up + 80-min match + half-time) (one energy gel, 0.75 L sports drink, 30 g trail mix, 0.5 L water).*		*Resistance training and light mobility (1 hr) 0.75 L water throughout training.*
Meal 3 (17:30 p.m.)	160 g white rice, ½ cup stir-fry vegetables pan-fried in olive oil, 140 g extra lean beef mince, stir-fry sauce, and 80 g broccoli. 1.0 L water.	Demi-baguette with 110 g chicken (deli meat), 100 g mixed salad, and 1 tbsp barbecue sauce. Cereal bar and one medium apple. 0.5 L water.	One medium (grilled) skinless chicken breast, one large baked sweet potato, 1 cup vegetable stir-fry (fried in olive oil), and 1 × 75 g focaccia herb/garlic bread. 0.5 L water.
Meal 3 code	MOD	MOD	HIGHER
Meal 4 (18:00–21:00 p.m.)	Large bowl of cornflakes made with a tall glass (330 ml) semiskimmed milk and 1 tbsp sugar. Protein shake (¾ pint semi-skimmed milk and flavored whey protein).	Two tuna wraps, each with: wholemeal tortilla, 100 g mixed salad, 1 tbsp low-fat mayonnaise, 1.5 heaped tbsp sweet corn, and ½ can tuna; and Nakd bar. 0.5 L water.	Pineapple and mango smoothie—1 pint semiskimmed milk, one scoop whey protein, two slices fruited malt bread, 125 g frozen mango, 160 g frozen pineapple, and 30 g almonds with skin. Extra 0.5 L before bed.
Meal 4 code	LOWER	MOD	MOD
Meal 5 (21:00 p.m.)		Two slices brown toast with ½ can baked beans and 15 g cheddar cheese, 150 g plain, and low-fat Greek yogurt with two small oranges. 0.5 L water.	
Meal 5 code		LOWER	
Meal 6 (22:00 p.m.)		125 g low fat, plain yogurt with 21 g topping of blueberries and 2 × 250 ml cups of milk.	
Meal 6 code		LOWER	
Daily nutrition profile	kcal: 3,422 (RMR = 2,436 × PAL 1.4 = 3,410) CHO 5.2 $g^{-1}\cdot kg^{-1}$, PRO 1.8 $g^{-1}\cdot kg^{-1}$, FAT 1.0 $g^{-1}\cdot kg^{-1}$ (478 g) (170 g) (92 g)	kcal: 4,221 (RMR = 3,007 × PAL 1.4 = 4,210) CHO 6.1 $g^{-1}\cdot kg^{-1}$, PRO 2.6 $g^{-1}\cdot kg^{-1}$, FAT 1.2 $g^{-1}\cdot kg^{-1}$ (562 g) (244 g) (111 g)	kcal: 3,376 (RMR 2,782 × PAL 1.4 = 3,357) CHO 4.5 $g^{-1}\cdot kg^{-1}$, PRO 2.0 $g^{-1}\cdot kg^{-1}$, FAT 1.2 $g^{-1}\cdot kg^{-1}$ (415 g) (183 g) (109 g)

Key		Meal code	Carbohydrate content
MD 0	Match day	HIGHER	Higher: 1.2–1.5+ g·kg
MD +1	The day after match day	MOD	Moderate: 1.0–1.2 g·kg
MD 2	Two days after match day	LOWER	Lower: 0.5–1.0 g·kg

Note. These values are therefore meant to represent the bare minimum requirements for health and performance. Meals highlighted in **bold** show examples of the nutritional contents of meals. Fluid requirements may fluctuate depending on training responses, for example, sweat rate. Differences in meals and codes account for appetite variation. Estimated using Nutritics (2019). CHO = carbohydrate; FAT = dietary fat; PAL = physical activity level; PM = afternoon; PRO = protein; RL = rugby league; RMR = resting metabolic rate; tbsp = tablespoon; tsp = teaspoon.

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Table 5 A Similar Diet for the Same for a 16- to 18-Year-Old, 92.4-kg Male RL Player Aiming to *Gain* Body Mass In-Season Based on the Estimated RMR of Costello et al. (2019) With Physical Activity Levels of 1.4 as "Light" (e.g., Sedentary/Seated Work; Black, 2001)

Diet and activity	MD 0	MD +1	MD +2
Meal 1 (07:30 a.m.)	Omelette (two eggs), 85 g spinach, and one medium tomato. Two crumpets (1 tsp butter each) and 140 g fruit salad. 0.5 L water.	One pack Belvita fruit and fiber biscuits, omelette (two eggs), 80 g spinach, and one medium tomato. Two crumpets (1 tsp butter each) and 140 g fruit salad. 0.5 L water. Extra 0.5 L water throughout morning.	Very large bowl muesli (95 g) made with 350 ml semiskimmed milk, 3½ tsp sunflower and pumpkin seed mix, and two large oranges. 0.5 L water. Extra 0.5 L water throughout morning.
Meal 1 code	LOWER	MOD	MOD
AM training (09:30 a.m.)			*Light rugby skills/technical (1 hr) 0.75 L water throughout training.*
Meal 2 (11:00 a.m.)	Two sandwiches, each with: two slices deli turkey, ½ salad tomato, small handful lettuce, and 1 tsp unsalted butter. 0.5 L water.	Two sandwiches, each with: two slices deli turkey, ½ salad tomato, small handful lettuce, and 1 tsp unsalted butter. 0.5 L water.	Smoothie—750 ml skimmed milk, 2 tbsp honey, 85 g blueberries, 85 g mango, 1 cup sour cherries, and 1 tsp hazelnut butter with a cereal bar.
Meal 2 code	LOWER	LOWER	HIGHER
Meal 3 (13:30 p.m.)	Large (426 g) Tuscan pasta bake and 80 g courgette with 250 ml fruit juice. 0.75 L water.	Large (426 g) Tuscan pasta bake and 80 g courgette and one can chickpeas with 250 ml fruit juice. 0.75 L water.	Two wholemeal pitas, each with: 1.5 tbsp hummus, ¾ cup sliced cucumber, and ½ cup grated carrot; and a Nakd bar. 0.75 L water.
Meal 3 code	MOD	MOD	MOD
PM training/match (15:00 p.m.)	*Match (warm-up + 80 min match + half-time) (600 ml apple juice + pinch of salt, one large banana, one Nutri-Grain fruit breakfast bar, 0.5 L water).*		
Meal 4 (17:30 p.m.)	160 g medium portion easy-cook brown rice (cooked), 160 g chicken curry with vegetables and curry sauce, 80 g bell pepper, 80 g spring onions, and ½ plain paratha flatbread. 0.75 L water.	300 g large portion easy-cook brown rice (cooked), 160 g chicken curry with vegetables and curry sauce, 80 g bell pepper, 80 g spring onions, and ½ plain paratha flatbread. 0.75 L water.	Medium portion prawns with 280 g egg noodles and Szechuan sauce and a side of Vietnamese rice paper rolls with hoisin sauce. 0.75 L water.
Meal 4 code	MOD	HIGHER	HIGHER (assuming heavy day the next day)
Meal 5 (20:30 p.m.)	Two white bagels, each with: 112 g smoked salmon, 1 tbsp extra light cream cheese, and ¾ cup cucumber slices; and Nakd bar. 0.75 L water.	One white bagel with 112 g smoked salmon, 2 tbsp extra light cream cheese, ¾ cup cucumber slices, Nakd bar, one pack carrot sticks, and hummus. 0.75 L water.	**Five rice cakes with 1 tbsp low-fat cottage cheese each and a Clif Peanut Butter and Chocolate energy bar. Dessert: 1 cup strawberries and 75 g chocolate ice cream. 0.5 L water. kcal: 1,079 39 g PRO (0.4 $g^{-1}\cdot kg^{-1}$) 108 g CHO (1.2 $g^{-1}\cdot kg^{-1}$) 23 g FAT (0.2 $g^{-1}\cdot kg^{-1}$)**
Meal 5 code	HIGHER	MOD	HIGHER
Meal 6 (22:00 p.m.)		Smoothie bowl—1 pint semiskimmed milk, 2 tbsp peanut butter, 250 g quark, 1 cup strawberry halves, and 4 tbsp honey. 0.5 L water.	
Meal 6 code		HIGHER	
Daily nutrition profile	kcal: 3,945 (RMR 2,436 × PAL 1.4 = 3,410 + 500 = 3,910) CHO 5.6 $g^{-1}\cdot kg^{-1}$, PRO 2.1 $g^{-1}\cdot kg^{-1}$, FAT 1.3 $g^{-1}\cdot kg^{-1}$ (521 g) (191 g) (118 g)	kcal: 4,713 (RMR 3,007 × PAL 1.4 = 4,210 + 500 = 4,710) CHO 6.3 $g^{-1}\cdot kg^{-1}$, PRO 2.6 $g^{-1}\cdot kg^{-1}$, FAT 1.6 $g^{-1}\cdot kg^{-1}$ (586 g) (242 g) (151 g)	kcal: 4,397 RMR 2,782 × PAL 1.4 = 3,895 + 500 = 4,395) CHO 7.3 $g^{-1}\cdot kg^{-1}$, PRO 1.9 $g^{-1}\cdot kg^{-1}$, FAT 1.2 $g^{-1}\cdot kg^{-1}$ (678 g) (179 g) (108 g)

Key		Meal code	Carbohydrate content
MD 0	Match day	HIGHER	Higher: 1.2–1.5+ g·kg
MD +1	The day after match day	MOD	Moderate: 1.0–1.2 g·kg
MD 2	Two days after match day	LOWER	Lower: 0.5–1.0 g·kg

Note. These values are therefore meant to represent the bare minimum requirements for health and requirements, considering varying appetites. Meals highlighted in **bold** show examples of the nutritional contents of meals. Fluid requirements may fluctuate depending on training responses, for example, sweat rate. Differences in meals and codes account for appetite variation. Estimated using Nutritics (2019). AM = morning; CHO = carbohydrate; FAT = dietary fat; PAL = physical activity level; PM = afternoon; PRO = protein; RL = rugby league; RMR = resting metabolic rate; tbsp = tablespoon; tsp = teaspoon.

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Table 6 Daily Carbohydrate Periodization for a 104-kg (Mean Age; 25.7 ± 4.1 Years) Player (Hudson et al., 2020) Aiming to Meet 6–8 g/kg CHO

Diet and activity	6 g/kg	7 g/kg	8 g/kg
Meal 1 (07:30 a.m.)	Large bowl porridge (80 g dry oats) made with 375 mL semiskimmed milk, one medium banana, 3 tbsp honey, and 125 g 0% fat plain yogurt. 40 g trail mix. 0.5 L water. Extra 1.0 L water throughout morning.	Three larger protein pancakes made with two scoops whey protein, one medium banana, 5 tbsp honey, and 150 g whole Greek yogurt. 1.5 L throughout morning.	**One tin of baked beans and three scrambled eggs on two thick slices of buttered brown toast, one tall tumbler of orange juice, and one large apple. 1.0 L water throughout morning.** **kcal: 1,422** **52 g PRO (0.5 $g^{-1}\cdot kg^{-1}$)** **150 g CHO (1.4 $g^{-1}\cdot kg^{-1}$)** **68 g FAT (0.7 $g^{-1}\cdot kg^{-1}$)**
Meal 1 code	HIGHER	HIGHER	HIGHER
Meal 2 (12:30 p.m.)	Smoothie—1 pint semiskimmed milk, one scoop whey protein, 1 tbsp peanut butter, 1 cup strawberries, 3 tbsp honey, Nakd bar, 30 g almonds, and 1 cup chopped dates. 0.5 L water before next meal.	Two ham and cheese paninis, 4 tbsp pasta salad with a large mixed garden salad, two Nature Valley protein bars, small bunch of grapes, and a large cappuccino. 0.75 L water.	Two large bacon sandwiches in buttered brown rolls and a protein shake (one scoop whey protein, 1 pint semiskimmed milk, and 1 cup each of mango and strawberry halves). 1.0 L before next meal.
Meal 2 code	HIGHER	MOD	HIGHER
Meal 3 (15:30 p.m.)	**Medium portion (160 g) white rice, 80 g stir-fry vegetables pan-fried in olive oil, medium portion (140 g) extra lean beef mince with one tin chopped tomatoes, 250 g Chinese stir-fry sauce, and large portion broccoli (110 g). One pack of Nature Valley Oat bars. 0.75 L water.** **kcal: 10,667** **52 g PRO (0.5 $g^{-1}\cdot kg^{-1}$)** **156 g CHO (1.5 $g^{-1}\cdot kg^{-1}$)** **26 g FAT (0.3 $g^{-1}\cdot kg^{-1}$)**	One large baked potato, one tin of baked beans, 90 g cheddar cheese, two buttered Soreen malt loaf slices. 1.0 L water. Extra 1.0 L water before next meal.	Two egg mayo sandwiches on white bread; two slices banana bread, each with 1 tbsp apple and berry compote; and 300 ml apple juice. 1.0 L water before next meal.
Meal 3 code	HIGHER	HIGHER	HIGHER
Meal 4 (19:00 p.m.)	Cheesecake and mixed berries with 500 ml fruit juice. 0.5 L water before bed.	**Two thick slices of marmite on toast, one 426-mL bottle of Yazoo, and 1 cup mixed berries with 50 g dark chocolate (e.g., melted). 0.5 L water.** **kcal: 863** **35 g PRO (0.3 $g^{-1}\cdot kg^{-1}$)** **113 g CHO (1.1 $g^{-1}\cdot kg^{-1}$)** **30 g FAT (0.3 $g^{-1}\cdot kg^{-1}$)**	One pack microwaveable brown rice, ½ jar sweet and sour sauce, one medium chicken fillet and 1 cup mixed vegetables. Snack of five dark chocolate rice cakes. 1.5 L water before next meal.
Meal 4 code	MOD	MOD	HIGHER
Meal 5 (22:00 p.m.)		Trek Berry Burst energy bar. 0.5 L water before bed.	260 g each salmon nigiri and vegetable sushi, 150 g rainbow salad. 1.5 L water before bed.
Meal 5 code		LOWER	HIGHER
Daily nutrition profile	kcal: 4,115 CHO 6.0 $g^{-1}\cdot kg^{-1}$, PRO 1.1 $g^{-1}\cdot kg^{-1}$, FAT 1.0 $g^{-1}\cdot kg^{-1}$ (627 g) (167 g) (109 g)	kcal: 5,832 CHO 7.0 $g^{-1}\cdot kg^{-1}$, PRO 1.6 $g^{-1}\cdot kg^{-1}$, FAT 2.0 $g^{-1}\cdot kg^{-1}$ (729 g) (272 g) (203 g)	kcal: 6,673 CHO 8.0 $g^{-1}\cdot kg^{-1}$, PRO 2.7 $g^{-1}\cdot kg^{-1}$, FAT 2.3 $g^{-1}\cdot kg^{-1}$ (834 g) (284 g) (244 g)

Key	
Meal code	Carbohydrate content
HIGHER	Higher: 1.2–1.5+ g·kg
MOD	Moderate: 1.0–1.2 g·kg
LOWER	Lower: 0.5–1.0 g·kg

Note. Meeting these recommendations may be challenging irrespective of training schedules, particularly for players wanting to lose weight, as in Table 2. For example, 6,673 kcal (8 g/kg) would require an average of 1,335 kcal per meal, for five meals. Meals highlighted in **bold** show examples of the nutritional contents of meals. Fluid requirements may fluctuate depending on training responses, for example, sweat rate. Differences in meals and codes account for appetite variation. Estimated using Nutritics (2019). CHO = carbohydrate; FAT = dietary fat; PAL = physical activity level; PM = afternoon; PRO = protein; RMR = resting metabolic rate; tbsp = tablespoon.

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increased energy demands for growth and maturation (Smith et al., 2018) and may benefit from focusing on the recommendations of "fuel for the work required" by ensuring key training sessions and competition are undertaken with high CHO availability to promote performance and recovery (Impey et al., 2018). The preparatory phase intervention of Morgan and Callister (2011) saw RL players self-manage body composition manipulations, losing approximately 0.6–1.1 kg fat and gaining approximately 0.7–0.8 kg muscle mass. Dietary intakes were not assessed, although reducing CHO intake to 4–6 $g \cdot kg^{-1} \cdot day^{-1}$ would be consistent with the consensus for a nutritionally adequate diet plan (Sundgot-Borgen & Garthe, 2011). Conversely, an intake of approximately 5–7 $g \cdot kg^{-1} \cdot day^{-1}$ CHO supported a weekly weight gain of ~0.5 kg in an RL player (Costello, Mckenna et al., 2018), gaining 4.7 kg muscle mass. Research is required to identify the potential ranges of CHO intake for optimal health and performance in RL squads. Recovery, and, hence, optimal nutrition, is vital during the competitive phase; thus, players likely require 5–7 $g \cdot kg^{-1} \cdot day^{-1}$ CHO for moderate activities (Thomas et al., 2016), whereas intense training (3–6 hr, 1–2 sessions daily) requires 8–10 $g \cdot kg^{-1} \cdot day^{-1}$ (Kerksick et al., 2018). Players should target high CHO availability for key training sessions and matches, possibly using "red, amber, or green" ratings corresponding to high, medium, and low CHO intake with flexibility to athlete history, training status, and specific training goals (Impey et al., 2018). Professional RL players can utilize ~40% of their muscle glycogen during a competitive match regardless of whether they consume 3.0 or 6.0 $g \cdot kg \cdot day^{-1}$ in the preceding 36 hr (Bradley et al., 2016). It was proposed that consuming ~600 g in the 36 hr before a match could ensure that all players commence a match with appropriate muscle glycogen concentrations (Bradley et al., 2016). Future research should establish whether players can consistently consume ~600 g CHO in the 36 hr before matches, particularly when traveling for away games and training camps. Players can meet their CHO intake goals using a mix-and-match strategy determined by personal preference (Jeukendrup, 2014).

Posttraining Requirements

The amount of CHO intake required postexercise depends on whether rapid restoration of muscle glycogen is the priority (Burke et al., 2017). After a simulated RL exercise protocol, immediate (1 hr post) CHO intakes of 1.0 g/kg *very likely* increased glycogen (53%), compared with delayed refeeding (2 hr post) (27%) (Bradley et al., 2017), although it is difficult to draw conclusions from 15 university-standard players. Consuming foods within 1 hr should suit players who lack an appetite immediately postexercise (Bradley et al., 2017), depending on preferences. During the preparatory phase, aggressive CHO refeeding of 1.0–1.2 $g \cdot kg^{-1} \cdot hr^{-1}$ for the first 4 hr may be useful when recovery between sessions is less than 8 hr (Burke et al., 2011). Alternatively, an initial feed of 1.0–1.2 g/kg CHO within 1 hr following training plus 0.6–1.0 g/kg every 2 hr for 6 hr (Thomas et al., 2016) may suit. Given the high energy demands of RL players (Morehen et al., 2016), consuming 1.2 g/kg CHO during the early stages of postexercise recovery can accelerate muscle glycogen repletion, beyond which glucose plus fructose or sucrose appears to be better tolerated than glucose alone (Gonzalez et al., 2016). Consuming glucose or galactose with fructose could double liver glycogen repletion rates (Gonzalez et al., 2016), while sucrose consumption (combining glucose and fructose polymers) appears to prevent liver, but not muscle, glycogen depletion (Gonzalez et al., 2015). This may explain the recent finding that after glycogen-depleting exercise, consuming 1.5 g/kg of glucose plus fructose in 20% solutions every 30 min during recovery improved performance in a time-to-exhaustion test 4 hr later by 27.2 ± 27.5% compared with glucose alone (Gray et al., 2020). An additional benefit to multiple transportable CHO consumption is reduced gastrointestinal distress. Following glycogen-depleting exercise, trained cyclists presented lower subjective ratings of upper gastrointestinal distress after consuming 1.5 g/kg of sucrose, a multiple-transportable CHO source, than with glucose ingestion (Fuchs et al., 2016). The authors explained that this was likely due to more transporters in the gastrointestinal tract being utilized after consuming multiple-transportable CHO, reducing water retention, improving absorption, and ultimately alleviating upper abdominal discomfort. During early recovery from training, within 1 hr, if possible, absolute intakes of 100–180 g CHO are recommended for RL players weighing approximately 84–120 kg. Presenting a "nutrition table" in dressing rooms with appealing foods, such as fruits and juices, cereal bars, and milk drinks and derivatives, could encourage adequate consumption. The evening meal should include CHO-rich foods, such as those shown in Tables 2 and 5.

Protein

Preparatory and Competition Phase Requirements

Consuming 1.4–2.0 $g \cdot kg^{-1} \cdot day^{-1}$ protein will support metabolic adaptation, repair, and remodeling (Thomas et al., 2016). Research suggests that 20 g of high biological value protein ingested soon after resistance exercise will maximally stimulate MPS (Slater & Phillips, 2011), although 40 g may be better for resistance-trained men (Macnaughton et al., 2016). Assuming players are resistance-trained (Schoenfeld & Aragon, 2018), consuming 20–40 g protein every 3–4 hr, with 6 g of essential amino acids, will maximize MPS (Jäger et al., 2017). Appreciating the limitations of self-report data, reported intakes of 2.2 $g \cdot kg^{-1} \cdot day^{-1}$ (Tooley et al., 2015) to 2.6 $g \cdot kg^{-1} \cdot day^{-1}$ (Costello et al., 2019) suggest that RL players may consume adequate protein; thus, players should ensure that high intakes come from practical, high-quality sources, such as biltong, chicken breast, eggs, and yogurt, as seen in the case study of Costello, McKenna et al. (2018).

Posttraining Requirements

Young, professional collision-sport athletes should fuel for the "muscle damage caused" as well as the kinematic "work required" (Costello et al., 2018a). Consuming protein as soon as possible following training is recommended to maximize recovery, with skeletal muscle remaining sensitive to protein ingestion for ~24 hr (Morton et al., 2015). Total daily intakes should therefore be prioritized over timing (Kerksick et al., 2017). However, consumption within 1 hr following a resistance or collision/contact training session is encouraged (Bradley et al., 2017). Alternatively, combining ~0.4 g/kg protein and 0.8–1.0 g/kg CHO supports glycogen recovery in RL players (Bradley et al., 2017), and when 20 g protein was co-ingested with 45 g CHO, postexercise MPS rates were shown to be similar between protein sources (whey, soy, and leucine-enriched soy) during recovery from concurrent exercise in young, recreationally active men (Churchward-Venne et al., 2019). However, assuming glycogen depletion, the rate of muscle glycogen resynthesis depends on the ingestion of CHO within the first 8 hr of recovery (Betts & Williams, 2010). Presleep protein consumption improves whole-body protein net balance overnight,

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with 35 g casein potentially increasing muscle mass and strength irrespective of timing (Joy et al., 2018). Intakes of 0.25–0.30 g/kg protein per meal favor recovery (Thomas et al., 2016), although adolescent RL players possibly consume up to 0.7 g/kg postexercise (Thivel et al., 2015), perhaps unnecessarily. Approximately 0.2 ± 0.6 g/kg or 0.25 ± 0.13 $g \cdot kg^{-1}$·lean body mass $(LBM)^{-1}$ (40–60 g for a 100-kg player) may be required to maximize MPS in younger men (Moore et al., 2014). Tables 2–6 show examples of high-quality protein sources, such as beef mince, tuna, and protein supplements containing at least 20 g high biological value protein, as reported previously in young rugby players (MacKenzie-Shalders et al., 2016).

Fluid

Preparatory and Competition Phase Requirements

During training and competition, athletes need to avoid fluid losses that exceed 2% BM (i.e., players weighing 80–100 kg should not lose >1.6–2.0/kg BM) to prevent dehydration, which can increase the perception of effort and impair exercise performance (Thomas et al., 2016). Players weighing approximately 83–112 kg have been known to average 1.28 ± 0.7 kg BM loss (1.31% BM) per match during a Super League season, with large variability of 0.16–2.74 kg (0.16–2.65% BM) (O'Hara et al., 2010). Research is required to determine if such findings are representative of the modern game. Consuming fluids periodically during exercise can help minimize losses, and improving fluid accessibility and the palatability of drinks may encourage adequate consumption during recovery. Adding sodium to fluids at a concentration equal to or greater than that of the sweat lost (20–80 mmol/L) is required to maintain positive fluid balance and is somewhat higher than the concentrations typically seen in commercial sports drinks (approximately 20–25 mmol/L; Evans et al., 2017). Offering players' preferred sports drinks, chilled and flavored for palatability, during half-time in matches will help maintain hydration. Postexercise, players should consume approximately 1.25–1.5 $L \cdot kg^{-1} \cdot BM^{-1}$ lost (Sawka et al., 2007) to ensure adequate rehydration.

Diet Quality

It is important to consider that reference dietary intakes fail to account for intense physical activity in athletes. For example, dietary reference intakes in the United States typically underestimate athletes' energy requirements (Thomas et al., 2016). Furthermore, reference nutrient intakes in the guidelines of the United Kingdom were defined as "two notional standard deviations above the estimated average requirement" (Department of Health, 1991). Reference nutrient intakes were said to be appropriate for the general population, yet an individual consuming a nutrient at levels that meet the reference nutrient intakes could still be consuming insufficient amounts of that nutrient (Department of Health, 1991). Adolescent rugby players have been reported to meet the U.K. dietary recommendations for energy, macronutrient, and fluid intake, and should therefore prioritize improving diet quality by replacing fat and sugar intake with healthier choices (Smith et al., 2016). Calcium and vitamin D are important for bone health, while poor iron status and iron deficiency, with or without anemia, may impair muscle function and work capacity, compromising training adaptations and athletic performance (Thomas et al., 2016). Iron is said to be one of the most critical micronutrients in the field of exercise nutrition, although dietary iron availability may be reduced by impaired intestinal absorption or inadequate energy intake (Koehler et al., 2012). Achieving adequate daily dietary iron intake (i.e., ≥18 mg for women and ≥8 mg for men) can help maintain the body's iron status, particularly during periods of intense training where there is the increased potential for increased iron losses through sweat and intravascular hemolysis (Thomas et al., 2016). However, practitioners should be cognizant of the differences between iron status and iron intake and the factors that may influence iron status, such as the bioavailability of different dietary sources of iron, the presence or absence of simultaneous intake of promoters or inhibitors of absorption, and the individual variability in intestinal absorptive capacity. The available data suggest that professional RL athletes (age = 18.1–32.7 years) have been shown to consume sufficient amounts of iron (28.1 ± 9.4 mg) and calcium (1,654 ± 591 mg) on average but consume high levels of alcohol following competition (Lundy et al., 2006), despite evidence showing that it potentially compromises MPS (Parr et al., 2014) and impairs training adaptations (Duplanty et al., 2017) and recovery by reducing peak power and cognitive recovery the day after a match (Murphy et al., 2013). Comparing the percentage change in performance at 2 hr and 16 hr postmatch, Murphy et al. (2013) found that consuming 233 ± 29 mL of vodka, as opposed to an orange juice control condition, 4 hr after a match significantly reduced countermovement jump height (−10.5 ± 8.4% vs. −2.4 ± 8.1%) and slowed cognitive response times in a modified Stroop test (−4.4 ± 8.4% vs. 6.2 ± 10.4%) on average. Alcoholic beverages contained 37.5% and 1 g/kg·BM of alcohol per volume, which the authors noted was equivalent to 7.7 ± 0.9 × 30 ml drinks. Therefore, similar intakes may be detrimental and should be avoided postmatch. Dietary fat provides energy and facilitates the absorption of fat-soluble vitamins, accounting for 20–35% of total daily energy intake in most athletes, which should be individualized based on training, body composition goals, and public health guidelines (e.g., <10% saturated fat; Thomas et al., 2016).

Performance Supplements

Nutritional supplements may help correct ongoing dietary deficiencies following clinical observations and dietary interventions (Larson-Meyer et al., 2018). An evidence-informed risk-benefit analysis should precede the consideration of supplementation. However, even with adequate energy intake and a balanced diet, some supplements can enhance recovery, adaptations to training, and performance during the precompetition and competition phases (Heaton et al., 2017).

Preparatory and Competition Phase Strategies

The effectiveness of potentially ergogenic supplements depends on the dosage, timing, type, exercise context, individual goal, and clinical need, and their consideration should be tailored accordingly. Specific to RL, the efficacy of creatine monohydrate creatine monohydrate (CM) (Cook et al., 2011; Kreider et al., 2017), caffeine (Del Coso et al., 2013; Wellington et al., 2017), beta-alanine (β-alanine) (Minett et al., 2010; Smith et al., 2019), and polyphenol (Bell et al., 2016; Howatson et al., 2012) supplementation is summarized below, excluding macronutrients.

Creatine monohydrate. Baseline phosphocreatine concentrations significantly influence power output and sprint/strength performance (Kreider et al., 2017); thus, CM supplementation likely improves these outcomes, as seen in young healthy males

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(Vandenberghe et al., 1999). However, there are potential responders and non-responders to ergogenic effects, with this research highlighting performance improvements only when a ≥20 mmol·kg^{-1}·day^{-1} increase in muscle total creatine occurred (Vandenberghe et al., 1999), which is immeasurable in practice. Research has also found that CM supplementation during periods of resistance training can result in greater gains in muscle mass (Kelly & Jenkins, 1998), and RL players may benefit from supplementing with CM during intense training blocks to improve recovery between high-intensity bouts within training sessions.

Relevance to RL: Though there has not been research on CM supplementation in RL players, relatable studies have been conducted in RU players (Chilibeck et al., 2007). Muscular endurance, measured as the maximum number of repetitions performed at 75% one-repetition maximum bench and leg press, improved following 8 weeks of 0.1 g·kg^{-1}·day^{-1} (averaging ~9.5 g) CM supplementation compared with a placebo group. Those receiving CM (+5.8 ± 1.4 repetitions) had a greater increase in the number of repetitions for the two muscular endurance tests combined compared with the placebo group (+0.9 ± 2.0). In addition, time main effects for BM (–0.7 ± 0.4 kg) and fat mass (–1.9 ± 0.8 kg) were observed, with a trend for increased lean tissue mass (+1.2 ± 0.5 kg) yet with no differences between groups. This led to the conclusions that CM supplementation during a RU season effectively increased muscular endurance but had no effect on body composition or aerobic endurance. High anaerobic endurance capacities and lean mass are important in RL (Johnston et al., 2014), and these findings therefore suggest potential crossover, though the between-group design of the study failed to account for interindividual responsiveness (Kreider et al., 2017). Elsewhere, a crossover-design experiment saw 0.05–0.10 g·kg^{-1}·day^{-1} (4.5–9.0 g) CM improve rugby-specific repeated passing performance in sleep-deprived rugby players ($p < .001$; Cook et al., 2011). Ten elite players completed 10 trials of a passing skill test (20 repeats per trial) with placebo, CM, and caffeine conditions. These were completed after adequate (7–9 hr) and deprived (3–5 hr) sleep conditions, comprising five separate trials each. On inspection of the data, 0.1 g·kg^{-1}·day^{-1} CM appeared to increase the number of successful passes from ~6.5 to ~9.0 on players' nondominant side, and approximately 7.0–7.5 to ~9.0–9.5 on their dominant side, respectively. Though these studies are in RU players, the movement patterns are comparable to those of RL players (Gabbett & Hulin, 2018), and the use of CM is likely to provide similar advantages.

Although the level of RL-specific evidence is low, data from studies in RU suggest that CM is a potential ergogenic aid (Dziedzic & Higham, 2014). Players may benefit from "loading" ~0.3 g·kg^{-1}·day^{-1} (approximately 25–30 g for 84–100 kg players) for 5–7 days to maximize muscle creatine rapidly, plus 3–5 g/day thereafter, or may alternatively consume 3–5 g/day consistently (Kreider et al., 2017). Weight gain, likely from water retention, has been the only consistent side effect and CM has been identified as the most effective ergogenic supplement for increasing high-intensity exercise capacity and LBM during training (Kreider et al., 2017).

Caffeine. Though several mechanisms have been proposed to explain caffeine's ergogenic potential, it is widely accepted that caffeine mainly acts as an adenosine receptor antagonist and central nervous system stimulant (Burke, 2008). Caffeine competes to block adenosine receptor binding sites, increasing neurotransmitter release, motor unit firing rates, and dopaminergic transmission, consequently increasing alertness and reducing fatigue and perception of effort (Burke, 2008). Consuming caffeine 1 hr preexercise ensures plasma concentrations peak at exercise onset and the half-life is approximately 2.5–10 hr (Goldstein et al., 2010). Players could ingest caffeine 30–60 min prematch to ensure peak concentrations are achieved during matches, with low to moderate dosages (approximately 3–6 mg/kg) conferring the greatest potential benefits (Goldstein et al., 2010).

Relevance to RL: The work of Wellington et al. (2017) showed that 300 mg caffeine (~3.4 mg·kg^{-1}·BM^{-1}) consumed 1 hr before exercise improved the time to complete nine 20-m sprints (28.46 ± 1.4 s) versus placebo (28.77 ± 1.7 s), with a *very likely* chance of benefit to repeated high-intensity effort ability (ES = 0.18, [–0.7, 0.1] s) in semiprofessional RL players. Moreover, improvements were evident earlier in the test, with time to complete 1.3% (ES = 0.19) and 1.0% (ES = 0.17) faster in Sets 1 and 2, respectively (*almost certain* benefits), and 0.9% (ES = 0.16) faster in Set 3 (*very likely* benefit) with caffeine. Supporting these findings, Del Coso et al. (2013) examined the effect of a caffeine-containing (6 mg·kg^{-1}·BM^{-1}) energy drink on simulated RU match performance, in which 30 players performed 2 × 30-min halves. Preexercise ingestion of an energy drink significantly increased total running distance, running distance at high intensities, and the number of body accelerations (tackles, changes of direction, and scrums).

The habitual caffeine consumption of RL players is unknown, and research must first understand dietary consumption before considering supplements. Caffeine is usually supplemented in 3–6 mg/kg dosages for ergogenic effects, with some rugby-specific evidence (Del Coso et al., 2013; Wellington et al., 2017) supporting the consensus that supplementation may benefit those in high-intensity intermittent team sports (Goldstein et al., 2010). Taken from said consensus, RL players may benefit from supplementing 3–6 mg/kg (approximately 240–600 mg for 80–100 kg players) approximately 1 hr prior to training or matches. Caffeine can be administered through gum, coffee, and energy drinks, although ergogenic effects appear stronger when consumed in tablets and powders (Goldstein et al., 2010), where a typical tablet may contain 100–200 mg (Burke, 2008). Coffee is the most popular caffeine source, and chronic caffeine intake does not appear problematic for most athletes, taking caution to possible side effects, such as sleep disturbances and anxiety, possibly influenced by genetic predispositions (Pickering & Kiely, 2019). Furthermore, habitual caffeine intake may deem supplementation unnecessary. Players should trial potentially effective strategies, considering preferences and the variability of caffeine in products (5–564 mg) and performance responses (Burke, 2008). To achieve potentially ergogenic dosages of 3–6 mg/kg, 80- to 120-kg players may benefit from supplementing lower dosages (e.g., 3 × 100 mg or 2 × 200 mg tablets) 1 hr preexercise.

Beta-alanine. Supplementation of β-alanine for 2–4 weeks increases muscle carnosine levels (4–6 g/day; Trexler et al., 2015). Muscle carnosine concentrations are related to the muscle's capacity to buffer changes in acid–base balance and therefore delay fatigue, particularly during high-intensity exercise. Intramuscular acidosis is most likely to occur during high-intensity exercise lasting 1–4 min, and given that RL relies on significant engagement of anaerobic glycolysis, β-alanine supplementation and the potential improvements in intramuscular buffering capacity may benefit RL players.

Relevance to RL: The prematch consumption of a multinutrient supplement, including 1 g β-alanine, 120 mg caffeine, and 1.5 g creatine malate, better maintained players' ability to complete very-high-speed running in the first half (5.9 ± 0.4 vs. 4.8 ± 2.3 m/min) and total distance in the second-half (137 ± 119 vs. 83 ± 89 m)

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of matches, compared with placebo, when taken during two separate 5-day periods, potentially due to the buffering properties of the ingredients (Minett et al., 2010). The contribution of β-alanine to these findings must be questioned as the 5-day protocol does not match the typical 2–4 weeks of supplementation recommended to increase muscle carnosine (Trexler et al., 2015). Multinutrient supplementation must be examined longitudinally in the competition phase to offer meaningful data. A recent pilot study showed no effect of 6.4 g/day β-alanine alone, compared with maltodextrin placebo, on resistance exercise performance in 15 male collegiate rugby players (Smith et al., 2019), though this held low statistical power and neglected the meaningfulness of rugby-specific performance outcomes. Having completed questionnaires to assess the prevalence of β-alanine supplementation-related knowledge and behaviors, approximately 80% of professional RU ($n = 87$) and RL ($n = 180$) players reported β-alanine use (Kelly et al., 2017), and most were not supplementing according to recommendations (4–6 g/day or 65–80 $mg \cdot kg^{-1} \cdot day^{-1}$) (Trexler et al., 2015). Acknowledging these factors, evidence suggests that 2–4 weeks of β-alanine supplementation (4–6 g/day) could improve exercise performance in 1- to 4-min tasks (Trexler et al., 2015). However, side effects include skin irritation, numbness, pins and needles/tingling, itchiness, and pain (Kelly et al., 2017), and the performance effects beyond 25 min and its applicability to RL require further investigation (Trexler et al., 2015).

Polyphenols. Fruit-derived polyphenols provide antioxidant and anti-inflammatory properties via mechanisms related to nitric oxide, such as production of superoxide and hence reduced conversion of nitric oxide to the cytotoxic oxidant, peroxynitrite, which, in turn, increases nitric oxide bioavailability (for review, see Bowtell & Kelly, 2019). Polyphenol supplementation may improve exercise performance given that excess exercise-induced reactive oxygen species generation could contribute to fatigue (Bowtell & Kelly, 2019), with a potential benefit to athletes who experience muscle damage as a result of explosive impacts, such as RL players (Costello et al., 2018b).

Relevance to RL: Antioxidant supplementation may benefit collision-sport athletes by targeting the secondary inflammatory response from impact-induced muscle damage (Naughton et al., 2018). In a performance context, Bell et al. (2016) randomly assigned 16 male soccer players to two groups and administered 60 ml/day Montmorency tart cherry concentrate or placebo for 8 days. Participants completed an adapted Loughborough Intermittent Shuttle Test on Day 5, and countermovement jump tests, a 20-m sprint (in seconds), and maximal voluntary contraction and muscle soreness ratings (200-mm visual analog scale; mm) were completed 24–72 hr postexercise. All performance measures recovered faster (e.g., 20-m sprint: 3.12 ± 0.15 s vs. 3.17 ± 0.19 s at 72 hr), and soreness was lower (e.g., 10 ± 8 mm vs. 33 ± 25 mm at 72 hr), with tart cherry compared with placebo. It was therefore suggested that tart cherry may accelerate recovery from prolonged, repeated sprint activity, as performed in RL. Elsewhere, research found that tart cherry did not affect muscle soreness, function, or inflammation following two matches in 11 professional RL players. Players participated in a single-blind randomized crossover design study, including habitual training and two consecutive scheduled matches during the competition phase of a season (Morehen, 2020). Two, separate 1-week supplementation periods included either tart cherry (2×320 mg of anthocyanins) or energy-matched placebo beverages. Subjective wellness (e.g., fatigue, sleep), soreness, and jump performance were measured pre- and postmatch, while GPS data estimated match demands. Furthermore, inflammatory markers were measured 48 hr prematch, at half-time, full-time, and 48 hr postmatch. Inflammatory markers were significantly increased at full-time compared with 48 hr pre- and postmatch (e.g., interleukin-6; 448% increase vs. 48 hr prematch); however, tart cherry had no significant effect. Unlike the aforementioned study (Bell et al., 2016), this research did not restrict dietary polyphenol intakes, and further research should seek to resolve these equivocal findings and investigate alternative benefits, such as improved sleep (Howatson et al., 2012).

Research must clarify RL players' habitual polyphenol-rich food intakes as, relevant to RL players, consuming ~300 mg polyphenols 1 hr preexercise may improve repeated sprint performance (Bowtell & Kelly, 2019) and recovery from prolonged, intermittent sprint activity (Bell et al., 2016). No adverse side effects have been reported, and practitioners may therefore focus on promoting compliance by individualizing protocols to players' preferences. Future research should identify the relative contributions and mechanisms of dietary polyphenols, melatonin, and CHO, which are often discussed in isolation (Howatson et al., 2012), or only briefly (Morehen, 2020), to recovery and performance outcomes.

Novelty Statement

The insights gained from this review may facilitate discussion of the energy, nutrient, supplement, and hydration demands across an RL macrocycle, providing initial evidence-based recommendations for the implementation of nutrient and supplement periodization for RL players.

Practical Application

Practical periodization strategies have been suggested in line with sport nutrition guidelines (Tables 2–6). Researchers should periodize nutrition to micro-, meso-, and macrocycles, allowing the creation of a rugby-specific framework. Individualization may supersede periodization, per se, given that during an RL season, players require constantly high intakes of energy, CHO, and protein to support the muscles' receptive environment for nutrients immediately postexercise and subsequently maximize performance and training adaptations.

Conclusions

The available data suggest that RL players have high energy requirements and must prepare and recover accordingly, carefully considering the multifactorial nature of recovery across a season to repeatedly optimize performance. The sport nutritionist/dietitian should synchronize with staff and players to periodize individual nutrition plans and group catering. The individualization of nutrition protocols, including efficacious supplements such as creatine, caffeine, β-alanine, and polyphenols, deserves exploration.

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